



B.Tech. Degree IV Semester Examination April 2014

ME 403 ADVANCED MECHANICS OF SOLIDS (2006 Scheme)

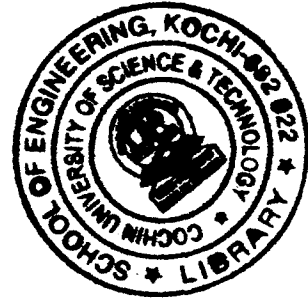
Time: 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) Explain the steps involved in the construction of Mohr's circle for strain measurement.
- (b) State and explain the compatibility equations for 2D problems.
- (c) Obtain the equation of equilibrium for rotating discs.
- (d) Draw the stress diagrams in composite tubes.
- (e) Obtain the differential equations of equilibrium of 3D problems.
- (f) Explain the term stress ellipsoid and stress invariants.
- (g) Obtain the expression for twist of a thin tube.
- (h) Explain membrane analogy.



PART B

(4 × 15 = 60)

- II. A material is subjected to two mutually perpendicular strains $\epsilon_x = 350 \times 10^{-6}$ units and $\epsilon_y = 50 \times 10^{-6}$ units together with an unknown shear strain γ_{xy} , if the principal strain in the material 420×10^{-6} units. Determine the following. (15)
 - (i) Magnitude of shear strain
 - (ii) The other principal strain
 - (iii) The direction of principal strain axes
 - (iv) The magnitude of the principal stresses
 If $E = 200 \text{ GN} / \text{m}^2$; $\nu = 0.3$.

OR

- III. For a rectangular rosette, the following observations are made with electrical strain gauges mounted on a steel specimen given (15)
 $E = 200 \text{ GPa}$, $\mu = 0.33$, $\epsilon_{0^\circ} = +400 \mu\text{m} / \text{m}$, $\epsilon_{45^\circ} = -200 \mu\text{m} / \text{m}$, $\epsilon_{90^\circ} = +200 \mu\text{m} / \text{m}$.
 Determine the principal strains, principal stresses and maximum shear stress.

- IV. (a) Derive an expression for radial and circumferential stress distribution for a thin, solid disc, which is rotating at an angular velocity ' ω '. (10)
- (b) An aluminium cylinder ($E_{al} = 0.7 \times 10^5 \text{ N} / \text{mm}^2$, $\nu_{al} = 0.33$) with an outer radius of 150mm and inner radius 100mm is to be press fitted over a steel cylinder ($E_{st} = 2 \times 10^5 \text{ N} / \text{mm}^2$, $\nu_{st} = 0.3$) with an outer radius of 100.25mm and inner radius 50mm. Determine the interference pressure. (5)

OR

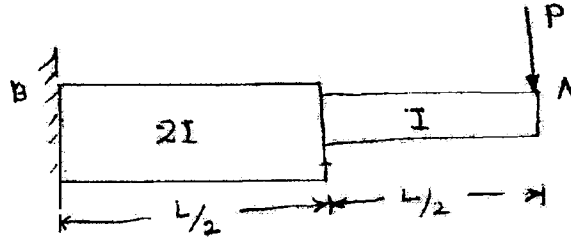
- V. Derive the equilibrium equations in polar co-ordinates. Also derive an expression for strain components in polar co-ordinates. (15)

(P.T.O.)

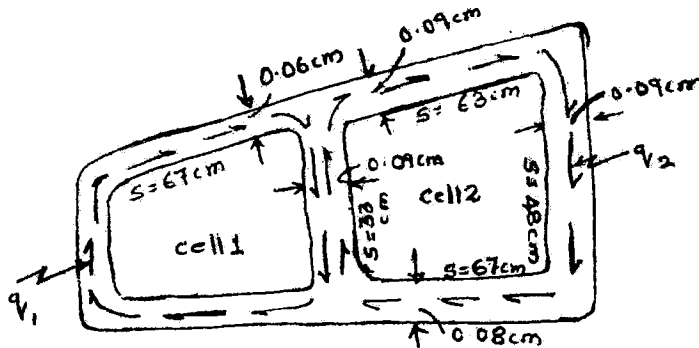
- VI. (a) Derive Cauchy's stress formulae. (5)
 (b) At a point P in a body, $\sigma_x = 10,000 \text{ N/mm}^2$, $\sigma_y = -5000 \text{ N/mm}^2$,
 $\sigma_z = -5000 \text{ N/mm}^2$, $\tau_{xy} = \tau_{yz} = \tau_{zx} = 10,000 \text{ N/mm}^2$. Determine the normal and shearing stresses in a plane that is equally inclined to all the three axes. (10)

OR

- VII. (a) Explain theorem of virtual work. (7)
 (b) A cantilever beam of stepwise constant cross section as shown below is loaded by a concentrated force at its tip. Determine the deflection under the point of application of the force by using Castigliano's theorem. (8)



- VIII. Figure shows a two cell tubular section as formed by a conventional air foil shape, and having one interior web. An external torque of $10,000 \text{ Nm}$ is acting in a clockwise direction. Determine the internal shear flow distribution. The cell areas are as follows. $A_1 = 680 \text{ cm}^2$, $A_2 = 2000 \text{ cm}^2$. The peripheral lengths are indicated in figure. (15)



OR

- IX. Determine the shear stress distribution in a channel section of a cantilever beam subjected to a load F. Also locate the shear centre. (15)

